Tara 11-11-26 Conversation 3: System diagram verification - laminate code, tape spring model and Abaqus comparison

Claude Code session, 11-11-2026, 17:16 - 18:40 PST. User messages and Claude replies in full; tool calls and outputs shortened; Claude's internal reasoning omitted.

User · 17:16

My goal in this session is to verify several steps in the system diagram in the Code for Project/11-4-26 (Tara)/system_diagram_verification folder. Please stand by for instructions

Claude · 17:16

Standing by. When you send the first step, I'll read the files in `Code for Project/11-4-26 (Tara)/system_diagram_verification` that it needs.

User · 17:20

I have a CFRP material with the following material properties: E1 = 139.5 GPA, E2 = 6.101 GPa, nu12 = 0.2750, G12 = 2.574 GPa, nu21 = 0.0120, and ply thickness is 54 microns. Use the laminate-properties skill to create the OpenSeesPy code for a [0/90/90/0] layup and convert that code to an ABD matrix, then output the ABD matrix here

update_plan task list

```
[in_progress] Load laminate-properties skill
[pending] Generate OpenSeesPy code for [0/90/90/0]
[pending] Extract ABD from OpenSees and compare with CLT
[pending] Report ABD matrix
```

```
Plan shown (4 steps).
```

Skill

```
{
 "skill": "laminate-properties",
 "args": "CFRP: E1=139.5 GPa, E2=6.101 GPa, nu12=0.2750, G12=2.574 GPa, nu21=0.0120, t_ply=0.054 mm; layup [0/90/90/0]; generate OpenSeesPy code and ABD matrix from it"
}
```

```
Launching skill: laminate-properties
```

Bash List verification folder and check python env

```
ls "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26 (Tara)/system_diagram_verification"; ls /c/Users/tvenkata/miniforge3/envs/shells/python.exe
```

```
ls: cannot access 'F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26 (Tara)/system_diagram_verification': No such file or directory
/c/Users/tvenkata/miniforge3/envs/shells/python.exe
```

Bash Find the verification folder

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project"; ls; ls -d */*erif* 2>/dev/null
```

```
10-22-26 (Federico)
10-23-26 (Tara)
10-26-26 (Federico and Tara)
10-26-26 (Federico)
10-29-26 (Federico)
11-06-26 (Orchestrator)
11-07-26 (Orchestrator)
11-10-26 (Tara)
11-11-26 (Tara)
11-4-26 (Tara)
[... shortened ...]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-4-26 (Tara)\system_diagram_verification.png

```
[image]
```

Bash Run laminate-properties for [0/90/90/0] with check and code emit

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-4-26 (Tara)" && mkdir -p
"system_diagram_verification/ply_layup_code_generator" && cd "system_diagram_verification/ply_layup_code_generator" &&
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK="F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills/laminate-
properties/laminate_properties.py"; $PY "$SK" --E1 139500 --E2 6101 --nu12 0.2750 --G12 2574 --angles 0 90 90 0 --thickness
0.054 --check --save-json lam_0_90_90_0.json --save-abd abd_clt.csv --emit-code opensees_laminate_0_90_90_0.py 2>&1; ls
```

```
Using ASDShellQ4 - Developed by: Massimo Petracca, Guido Camata, ASDEA Software Technology
Ply material: user ply  (N, mm, MPa)
E1          139500 MPa     input
E2            6101 MPa     input
E3            6101 MPa     = E2
G12           2574 MPa     input
G13           2574 MPa     = G12
G23        2392.55 MPa     = E2 / (2 (1 + nu23))
nu12          0.275        input
nu13          0.275        = nu12
[... shortened ...]
```

Claude · 17:20

Ran the skill; the CLT ABD matrix and the one measured from the OpenSeesPy code both came out. Results below.

wrap_up wrap-up

```
The laminate-properties skill produced the OpenSeesPy code for [0/90/90/0] and measured its ABD. A and B match CLT to about
1e-16 and D is 0.23% low, inside the expected bound. Nothing is committed.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 17:20

I generated the OpenSeesPy code for [0/90/90/0], measured its ABD matrix by imposing strains on one OpenSees shell element, and compared it with classical laminate theory (CLT). A and B match CLT to about 1e-16, and D is 0.23% lower, inside the 0.39% the skill expects for 16 layers.

ABD matrix measured from the OpenSeesPy code

Convention: N = A·ε + B·κ, M = B·ε + D·κ, with engineering γ_xy.

A [N/mm] | | ε_xx | ε_yy | γ_xy | |--------|---------|---------|-------| | N_xx | 15777.1 | 363.602 | 0 | | N_yy | 363.602 | 15777.1 | 0 | | N_xy | 0 | 0 | 555.984 |

B [N]: zero. The largest entry is 3e-13, which is round-off; the laminate is symmetric.

D [N·mm] | | κ_xx | κ_yy | κ_xy | |--------|---------|---------|-------| | M_xx | 103.252 | 1.40816 | 0 | | M_yy | 1.40816 | 18.9512 | 0 | | M_xy | 0 | 0 | 2.15322 |

Exact CLT values, for comparison

- **A:** the same as above.
- **B:** zero.
- **D:** D11 = 103.492, D12 = 1.41368, D22 = 19.1908, D66 = 2.16167.
- D11 also checks by hand: $2 \cdot Q11 \cdot (0.108^3 - 0.054^3)/3 + 2 \cdot Q22 \cdot 0.054^3/3 = 103.49$.

Why the OpenSees D is slightly low: OpenSees integrates each layer at its mid-plane. With 4 layers per ply (16 in total), that error is about $1/16^2$, so OpenSees comes out a little low.

Inputs and assumptions

- **Ply 1 is the bottom ply.** Each ply is 0.054 mm thick, so the laminate is h = 0.216 mm.
- **Units:** I entered E1 = 139500 MPa, E2 = 6101 MPa, G12 = 2574 MPa, nu12 = 0.275.
- **nu21:** the script computed it as nu12·E2/E1 = 0.012027, which matches your 0.0120.
- **nu23 was not given,** so nu23 = nu12 = 0.275 was assumed, which gives G23 = 2392.55 MPa. This does not affect ABD.
- **Matrix shape:**
- B = 0 because the laminate is symmetric.
- A16 = A26 = 0 because it is balanced.
- D16 = D26 = 0 because it has no ±θ plies, so there is no bend-twist coupling.
- **Coupled terms:** Q11 = 139963, Q22 = 6121.25, Q12 = 1683.34, Q66 = 2574 MPa.
- **Code structure:** the code uses two coincident ASDShellQ4 elements per facet, one for the 0° plies and one for the 90° plies, each with `-noeas`.

Files are in `Code for Project/11-4-26 (Tara)/system_diagram_verification/ply_layup_code_generator/`. The folder named in your first message doesn't exist (only `system_diagram_verification.png` does), so I created it. The measured OpenSees ABD is only in the console output above; `abd_clt.csv` holds the CLT values. - opensees_laminate_0_90_90_0.py: the OpenSeesPy material and section code - lam_0_90_90_0.json: the laminate definition, for the other skills - abd_clt.csv: the CLT ABD

User · 17:26

I want to re-create Federico's test of linear beam bending of an aluminum plate from 10/22 and comparison to beam theory, using the tools in their current state rather than the old version of agents/tools. Give me a prompt that will do so

Bash List Federico's 10-22 files

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/10-22-26 (Federico)" && find . -maxdepth 3 | head -60; ls ../../"Running Log" 2>/dev/null | head
```

```
.
./.gitignore
./elastica.py
./environment.yml
./flat_plate_al.py
./plate_large_deflection.py
./plate_tip_load.py
./README.md
./results
./results/large_deflection_ASDShellQ4-corot_10mm.csv
[... shortened ...]
```

Bash Read the 10-22 summary and linear test scripts

```
cat SUMMARY.md; echo ======; cat plate_tip_load.py | head -150; echo =====; head -60 flat_plate_al.py
```

```
# Summary of work: OpenSeesPy setup and shell-model verification

**Date:** 10/22/26
**Who:** Federico, working with Claude Code (Opus 5.5)
**Commit:** `b5e140c` on `main` ("Add OpenSeesPy thin-shell verification models")

## Why

Before an agent can explore the design space of thin composite shells, we need
three things: a working Python/OpenSeesPy environment on the AWS VM, a way to
[... shortened ...]
```

wrap_up wrap-up

```
I read Federico's 10-22 linear tip-load test (plate_tip_load.py, flat_plate_al.py, SUMMARY.md) and drafted a prompt that
rebuilds it with the current skills. Nothing has been run yet.
```

```
Nothing is pending, so the app draws no box; the recap is the session's status. Your closing message is your answer to the
user: write it now, as the last thing you say this turn.
```

Claude · 17:26

I based this prompt on Federico's `plate_tip_load.py`, `flat_plate_al.py` and `SUMMARY.md`. It uses the same inputs and the same checks as his test, but every step goes through the current skills:

```
Re-create Federico's 10/22 linear beam-bending test of an aluminum cantilever plate
("Code for Project/10-22-26 (Federico)/plate_tip_load.py" + flat_plate_al.py, results in
SUMMARY.md section 4) using ONLY the current skills in .claude/skills, not the old 10-22
scripts or the v1.0 agents. Read each SKILL.md before using it. Do not invoke design-boom
or the ThinShellOrchestrator. Use the shells env (C:\Users\tvenkata\miniforge3\envs\shells).
Units N, mm, MPa. Put all outputs in "Code for Project/11-11-26 (Tara)/al_plate_linear_bending/".

Model (same as 10/22):
- Flat aluminum plate, L = 1000 mm (along x), b = 100 mm, t = 0.1 mm;
  E = 69000 MPa, nu = 0.33, rho = 2.70e-9 t/mm^3 (isotropic shell).
- Clamped (all 6 DOF) on the short edge x = 0; free elsewhere.
- Total tip load P = -1e-5 N in z, spread as a uniform line load along the free edge x = L
  (consistent nodal loads), so the plate bends about its weak axis.
- Linear static analysis, 1 mm x 1 mm elements (100 x 1000 = 100,000 elements), as in 10/22.

Steps, one skill each:
1. laminate-properties: isotropic shell section for the aluminum (Laminate.isotropic); report it.
2. shell-mesh: flat strip 1000 x 100 mm, 1 mm elements; report element type, node/element count.
3. shell-bc-load: clamp at x = 0, edge line load at x = L, linear solve; save the results .npz.
4. shell-assemble-run: join the three pieces into ONE standalone OpenSeesPy script, run it,
   and keep the script and .npz in the output folder.
5. shell-visualize: PNG of the deformed plate over the undeformed outline, coloured by
   curvature and by top-surface sigma_xx.
6. Compare to Euler-Bernoulli beam theory with I = b t^3/12 (use the beam-theory reference
   in shell-bc-load where it exists, and state the formulas):
   - w(x) = P x^2 (3L - x) / (6 E I) on the mid-width line (y = b/2) at x = 250, 500, 750,
     1000 mm; tolerance 3%. Also report the edge-averaged tip deflection next to the
     wide-plate bound with E/(1 - nu^2).
   - top-surface sigma_xx(x) = -P (L - x)(t/2)/I at x = 250, 500, 750 mm on the mid-width
     row (element centroids, x + 0.5 mm); tolerance 3%. Report max |sigma_xx| at the root
     and explain the clamp/anticlastic effect.
7. Put a table side by side: current-tools result, beam theory, and Federico's 10/22
   result (SUMMARY.md: deflection within 1.2-2.8%, stress within 0.06%). Say PASS or FAIL for
   each check, and explain any difference from 10/22, e.g. element type (ShellMITC4 then
   vs. whatever the skills use now) or how the load and stress are recovered.

Write a short SUMMARY.md in the output folder (date, person, what was run, the table, the
files), and add a Running Log entry (date, person, entry). Don't commit until I review it.
```

- **Run time:** in 10/22 the 1 mm mesh took about 25 s to solve and had about 606k degrees of freedom. If the current skills run much slower, the prompt can allow a 5 mm mesh as a fallback, as long as the deviation is reported.
- **Lighter alternative:** the `regression-tests` skill already re-runs the verified 10-22 cases and compares them with v1. It's a quicker check, but it doesn't walk through each skill the way your system diagram does. The prompt above does, which is why I wrote it that way.

User · 17:32

I have a tape spring boom with the following properties: material L001 in the material property catalog, layup [0/0/0], radius of curvature 20 mm, subtended angle theta = 180 degrees, length 1 meter, cantilever boom clamped at 1 end. I want to understand how this boom will deform in beam bending under a 1 N tip load. Use the Python tools in .claude/skills (laminate-properties for material properties, shell-bc-load for mesh and boundary conditions, shell-assemble-run to create 1 cohesive OpenSeesPy code and run it, shell-visualize to visualize the results, and objective-constraints to evaluate the mass (which would be the objective function if I were running an optimization, which I'm not currently doing) and Hashin failure criteria)

update_plan task list

```
[in_progress] Read the skill docs and L001 entry
[pending] Build laminate, mesh, BC/load pieces
[pending] Assemble and run one OpenSeesPy script
[pending] Visualize the deformed boom
[pending] Evaluate mass and Hashin failure
[pending] Report results
```

```
Plan shown (6 steps).
```

Bash Check study state, list skills, find L001

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude; ls orchestrator/ACTIVE_STUDY 2>&1; ls skills/*/; wc -l skills/{shell-mesh,shell-bc-load,shell-assemble-run,shell-visualize,objective-constraints}/SKILL.md; grep -n "L001" memory.material_property_catalog/*.txt | head
```

```
ls: cannot access 'orchestrator/ACTIVE_STUDY': No such file or directory
skills/abaqus-export/:
SKILL.md
case_to_abaqus.py

skills/design-boom/:
SKILL.md

skills/failure-check/:
SKILL.md
[... shortened ...]
```

Bash Read shell-mesh and shell-bc-load skill docs

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills; cat shell-mesh/SKILL.md shell-bc-load/SKILL.md
```

```
---
name: shell-mesh
description: Builds an OpenSeesPy thin-shell component from a cross-section (flat strip, circular arc / tape spring, a chain of lines and arcs, any y-z polyline, or a branched / closed section such as a TRAC or omega / corrugated boom straight from the boom geometry catalog, with the doubled laminate on its bonded webs) extruded along x, meshed with 4-node shells of a given size, with the section from the laminate-properties skill and each facet's laminate orientation set from its own geometry. Use whenever a shell model's geometry and mesh are needed.
---

# shell-mesh

Script: `shell_mesh.py` (this folder; imports `../laminate-properties`). Units: N, mm, MPa.
The OpenSees domain only lives in the Python process that builds it, so this
skill is used from Python (a driver script or the shell-bc-load case runner),
[... shortened ...]
```

Bash Read assemble, visualize, objective-constraints docs

```
cat shell-assemble-run/SKILL.md shell-visualize/SKILL.md objective-constraints/SKILL.md
```

```
---
name: shell-assemble-run
description: Joins the OpenSeesPy code pieces written by the laminate-properties (materials and LayeredShell sections), shell-mesh (nodes and shell elements) and shell-bc-load (supports, loads, solution, results file) skills into ONE continuous, standalone OpenSeesPy Python script (needs only openseespy and numpy), checks that the pieces fit together, runs it and saves the results .npz for shell-visualize and failure-check. Use when the user wants a single self-contained
```

```
OpenSeesPy script of a shell model (to read, share, archive or run elsewhere), or wants the model assembled and run as one
script.
---

# shell-assemble-run

Script: `assemble_run.py` (this folder; imports `../laminate-properties`,
`../shell-mesh`, `../shell-bc-load`). Units: N, mm, MPa.
Python: conda env `shells` (openseespy 3.8): Federico
[... shortened ...]
```

Bash Find catalog column meanings and L001 ply thickness

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/memory.material_property_catalog; grep -n "^#.*ID\|^ID\|t_ply\|^# *col"
UD_lamina_catalog_SI.txt | head; sed -n 78,90p UD_lamina_catalog_SI.txt; grep -n "L001" UD_lamina_catalog_readable.txt |
tail -1; grep -n "^ID\|^#  *ID" UD_lamina_catalog_readable.txt
```

```
7:# Columns: ID | Fiber | Polymer system | E1 | E2 | G12 | nu12 | nu23 | G23 | Xc | Xt | Yc | Yt | S12 | t_ply | rho |
Source
14:#     t*    : see per-ID note.
18:# Per-ID notes
79:# PLY THICKNESS (project decision; 0.045 mm on 2026-11-04, 0.035 mm since 2026-11-05): t_ply = 0.035 mm for EVERY row, so
that all materials can be
116:ID     Fiber                          Polymer_system                                E1_Pa        E2_Pa       G12_Pa
nu12     nu23    G23_Pa      Xc_Pa         Xt_Pa         Yc_Pa        Yt_Pa       S12_Pa       t_ply_m   rho_kg/m3   Source
#
# PLY THICKNESS (project decision; 0.045 mm on 2026-11-04, 0.035 mm since 2026-11-05): t_ply = 0.035 mm for EVERY row, so
that all materials can be
#   used for thin-shell deployable structures, assuming they could be made as 35 um plies. Stiffness, strength and
#   density are still those measured at the original thickness. Thin plies usually show higher in-situ transverse and
#   shear strengths and somewhat different Xc, so the strengths here are an approximation for 35 um plies.
[... shortened ...]
```

Bash Write linear tape-spring case files at two mesh sizes

```
D="F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load"; mkdir -p "$D";
cd "$D"
for s in 5 2.5; do n=$( [ $s = 5 ] && echo h5 || echo h2p5 ); cat > ts_L001_000_linear_$n.json <<EOF
{
 "name": "ts_L001_000_linear_$n",
 "laminate": {"catalog": "L001", "layup": "[0/0/0]"},
 "geometry": {"L": 1000, "section": {"type": "arc", "R": 20, "b": 62.83185307179586}, "size": $s},
 "element": {"type": "ASDShellQ4", "corotational": false},
 "supports": [{"edge": "x0", "dofs": [1,1,1,1,1,1]}, {"rigid_end": "xL"}],
 "loads": [{"node": {"rigid": "xL"}, "force": [0, 0, -1], "moment": [0, 0, 0]}],
 "analysis": {"type": "linear"},
 "report_node": {"rigid": "xL"}
}
EOF
done; ls
```

```
ts_L001_000_linear_h2p5.json
ts_L001_000_linear_h5.json
```

Claude · 17:33

Case files written (L001, 3 × 0.035 mm plies, R = 20 mm, 180° arc, 1 m long, 1 N tip load). Next I'll run laminate-properties, then assemble and run the model.

Bash Build L001 [0/0/0] laminate with ABD check

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
$PY $SK/laminate-properties/laminate_properties.py --catalog L001 --layup "[0/0/0]" --check --save-json lam_L001_000.json
2>&1 | grep -v "^Using ASD"
```

```
Ply material: L001 IM7 / Hexcel HexPly 8552 epoxy (NCAMP)  (N, mm, MPa)
E1          158500 MPa     input
E2            8950 MPa     input
E3            8950 MPa     = E2
G12           4670 MPa     input
G13           4670 MPa     = G12
```

```
G23        3086.21 MPa     = E2 / (2 (1 + nu23))
nu12         0.316         input
nu13         0.316         = nu12
nu23          0.45         input
[... shortened ...]
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 6 is in progress.
```

Bash Assemble and run linear model at 5 and 2.5 mm meshes

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
for n in h5 h2p5; do echo "=== $n"; extra=$([ $n = h5 ] && echo --verify); time $PY $SK/shell-assemble-run/assemble_run.py
ts_L001_000_linear_$n.json --laminate lam_L001_000.json $extra 2>&1 | grep -v "^Using ASD"; done
```

```
=== h5
1000 x 62.8319 mm shell, arc R = 20 mm (180.0 deg); 200 x 13 = 2600 quads of 5 x 4.83 mm, 2814 nodes
2600 elements, 16890 DOFs; linear solved in 1.4 s
saved F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h5.npz
report node 2815: u = [0.0, 0.0, -8.596272, -0.0, 0.012773, -0.0]
max |u| = 8.59779 mm
laminate piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\ts_L001_000_linear_h5_code\laminate_code.py
    mesh piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\ts_L001_000_linear_h5_code\mesh_code.py
      bc piece: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\ts_L001_000_linear_h5_code\bc_code.py
joined script: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\ts_L001_000_linear_h5_opensees.py (5639 lines)
[... shortened ...]
```

Bash Compute beam-theory reference for the arc section

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
$PY - <<EOF 2>&1 | grep -v "^Using ASD"
import sys, numpy as np, inspect
sys.path[:0]=[r"$SK/laminate-properties", r"$SK/shell-mesh", r"$SK/shell-bc-load"]
import shell_mesh as sm, laminate_properties as lp
print(inspect.signature(sm.arc_section_inertia)); print(sm.arc_section_inertia.__doc__)
lam=lp.Laminate.load("lam_L001_000.json") if hasattr(lp.Laminate,'load') else None
A,B,D=[np.asarray(m) for m in lam.abd()] if lam else (None,)*3
a=np.linalg.inv(np.block([[A,B],[B,D]])); h=0.105
Ex=1/(a[0,0]*h); print("Ex_membrane",Ex)
r=sm.arc_section_inertia(20, np.pi*20, h) if 'h' in str(inspect.signature(sm.arc_section_inertia)) else
sm.arc_section_inertia(20, np.pi*20)
print(r)
EOF
```

```
(R, b, t, n=200001)
Thin-walled arc: centroid z_c and I about the horizontal centroidal
    axis (as verify_mesh_bc_load.py).
Traceback (most recent call last):
  File "<stdin>", line 6, in <module>
ValueError: too many values to unpack (expected 3)
```

Bash Compute beam-theory deflection, rotation, stress

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
$PY - <<EOF 2>&1 | grep -v "^Using ASD"
import sys, numpy as np
sys.path[:0]=[r"$SK/laminate-properties", r"$SK/shell-mesh", r"$SK/shell-bc-load"]
import shell_mesh as sm, laminate_properties as lp
lam=lp.laminate_from_spec({'catalog':'L001','layup':'[0/0/0]'})
abd=np.asarray(lam.abd()); print(abd.shape)
a=np.linalg.inv(abd); h=0.105; Ex=1/(a[0,0]*h)
zc,I=sm.arc_section_inertia(20, np.pi*20, h)
EI=Ex*I; L=1000; P=1
print(f"Ex={Ex:.1f} MPa zc={zc:.4f} I={I:.3f} mm^4 EI={EI:.4g} N mm^2")
```

```
print(f"PL^3/3EI = {P*L**3/(3*EI):.4f} mm, rot PL^2/2EI = {P*L**2/(2*EI):.5f} rad")
print(f"root M = {P*L} Nmm, sigma_max beam (edges, c={20-zc:.3f}) = {P*L*(20-zc)/I:.2f} MPa; bottom (c={zc:.3f})
{P*L*zc/I:.2f} MPa")
D=abd[3:,3:]; nu=0.316
print("D11",D[0,0],"D22",D[1,1])
EOF
```

```
(6, 6)
Ex=158500.0 MPa zc=7.2677 I=249.953 mm^4 EI=3.962e+07 N mm^2
PL^3/3EI = 8.4138 mm, rot PL^2/2EI = 0.01262 rad
root M = 1000 Nmm, sigma_max beam (edges, c=12.732) = 50.94 MPa; bottom (c=7.268) 29.08 MPa
D11 15.377000963222345 D22 0.8682912215825866
```

update_plan task list

```
{"done": [2, 3], "start": 4}
```

```
Plan updated: step 4 of 6 is in progress.
```

Bash Render deformed-shape PNGs for the 2.5 mm mesh

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
$PY $SK/shell-visualize/shell_visualize.py results/ts_L001_000_linear_h2p5.npz --png results/ts_L001_000_linear_h2p5.png --
no-gui --stress sigma_xx_top 2>&1 | grep -v "^Using ASD"
$PY $SK/shell-visualize/shell_visualize.py results/ts_L001_000_linear_h2p5.npz --png
results/ts_L001_000_linear_h2p5_hashin.png --no-gui --field hashin_max 2>&1 | grep -v "^Using ASD"
```

```
web facets: 0 of 10000 (no web (no catalog boom given))
saved results\ts_L001_000_linear_h2p5.vtp: 10000 facets, 10427 nodes
k_xx [1/mm]: min -0.00018868, max 0.00023911
sigma_xx_top [MPa]: min -29.601, max 46.142
saved results/ts_L001_000_linear_h2p5.png
web facets: 0 of 10000 (no web (no catalog boom given))
saved results\ts_L001_000_linear_h2p5.vtp: 10000 facets, 10427 nodes
hashin_max [-]: min 4.707e-08, max 0.00041716
saved results/ts_L001_000_linear_h2p5_hashin.png
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h2p5.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h2p5_hashin.png

```
[image]
```

Claude · 17:34

The linear model runs and looks right: tip deflection is 8.52 mm against 8.41 mm from beam theory. Next I'll compute the mass and Hashin failure with objective-constraints.

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 5 of 6 is in progress.
```

Bash Evaluate mass and Hashin failure for both meshes

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
for n in h2p5 h5; do echo "=== $n"; $PY $SK/objective-constraints/objective_constraints.py results/ts_L001_000_linear_$n.npz
2>&1 | grep -v "^Using ASD"; done
```

```
=== h2p5
Objective / constraints: ts_L001_000_linear_h2p5  [[0/0/0] h = 0.105 mm]
OBJECTIVE  minimize mass: 0.0104169 kg (10.42 g)
CONSTRAINTS (g <= 0 is satisfied; g_rel = g / |limit|)
  [satisfied    ] no failure: F_governing < 1.0  value 0.136288, limit 1, g -0.863712, g_rel -0.863712
FEASIBLE: yes
  note: catalog L001 rho = 1580 kg/m^3
  note: tip = free end xL: rigid end master node at xL
  note: S23 = Yc/2 (default)
  note: coiling: flange flattened from r = 20 mm (smallest arc radius of the case cross-section), F = 0.1363 (matrix
```

```
tension)
[... shortened ...]
```

wrap_up wrap-up

```
The linear model of the L001 [0/0/0] tape spring under a 1 N tip load ran on two meshes. Tip deflection is 8.52 mm (beam theory 8.41 mm), mass 10.4 g, and the governing Hashin F is 0.136, set by coiling. The linear run cannot show local folding, which a 1000 N·mm root moment probably causes. Nothing is committed.
```

```
Shown in the next-step box. Do not call wrap_up again this turn; the advice that follows is for your next call. recap is two or three sentences (about 300 characters); this one ran 314. Keep it to what is now true and where it stands. Your closing message is your answer to the user: write it now, in full, as the last thing you say this turn. The box asks your question, so do not ask it again.
```

Claude · 17:34

In a linear analysis this tape spring bends as a beam. It deflects 8.52 mm at the tip under 1 N, close to beam theory, and is far from failing. However, a linear analysis cannot show whether the thin section folds near the clamp, and at this load I think it probably does.

Model

- **Material:** L001 (IM7/8552), [0/0/0], 3 × 0.035 mm plies, so the laminate is 0.105 mm thick.
- The catalog's 35 µm ply thickness is a project assumption; the original NCAMP ply was 0.183 mm.
- nu23 = 0.45 and G23 come from the catalog and are estimates; they don't affect the ABD.
- The ABD measured from the OpenSees code matches CLT (A and B to about 1e-17; D is 0.44% low, as expected for 15 layers). D11 = 15.38 N·mm.
- **Geometry:** 180° arc with R = 20 mm (arc length b = 62.83 mm), 1000 mm long.
- The concave side faces +z, so the arc edges are at z = 20 mm and the centroid is at z = 7.27 mm.
- The mesh is made by the shell-mesh skill; shell-bc-load calls it from the case file. Element is ASDShellQ4, linear geometry.
- **Supports and load:** all 6 DOFs clamped at x = 0, with a rigid fixture at x = L.
- The load is 1 N in −z on that fixture, so the arc edges are in tension (opposite-sense bending).
- Loading in +z would put the edges in compression (equal-sense bending).

Results

	5 mm mesh (16,890 DOF)	2.5 mm mesh (62,562 DOF)	Beam theory
Tip deflection u_z	−8.596 mm	−8.523 mm	−8.414 mm ($PL^3/3EI$, EI = 3.96e7 N·mm^2)
Tip rotation	0.01277 rad	0.01266 rad	0.01262 rad
Max σ_xx at the root	46.1 MPa (2.5 mm mesh)		50.9 MPa at the edges (Mc/I)

- The 2.5 mm mesh is 1.3% more flexible than beam theory, and the two meshes differ by 0.9%.
- The single-script run gave exactly the same result as running the case directly (`--verify` PASS).

Mass and failure (objective-constraints)

- **Mass:** 10.42 g, using the catalog density of 1580 kg/m^3. A hand check (1000 × 62.83 × 0.105 mm^3 × 1.58 g/cm^3) gives the same.
- **Failure under the load:** the largest Hashin index is 4.2e-4. It is fibre tension in the bottom ply at the clamp, with σ1 ≈ 50 MPa against Xt = 2501 MPa.
- **Failure from coiling:** flattening the R = 20 mm arc gives F = 0.136 (matrix tension), and that sets the governing value. With F < 1 the design passes the no-failure check.

Why the linear result is probably not what happens

- **Root moment vs fold moment:** the root moment is 1 N × 1000 mm = 1000 N·mm. A rough estimate of the moment that keeps an open fold in this tape is about D11·θ ≈ 15 × π ≈ 50 N·mm.
- **Peak moment:** the moment needed to start a fold is several times that, but very likely still well below 1000 N·mm. That is an order-of-magnitude estimate; I haven't calculated it.
- **What that means:** the boom would probably flatten locally and fold near the clamp rather than bend smoothly. A linear analysis cannot capture this, so treat the 8.5 mm and the low Hashin index from the load as valid only before the fold forms. The coiling index doesn't depend on the run.
- **How to check:** a large-rotation (corotational) run with displacement control, or the shell-buckling skill, would show whether and where the fold forms.

Files are in `Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load/`: - ts_L001_000_linear_h2p5_opensees.py: the single OpenSeesPy script, 20,675 lines - ts_L001_000_linear_h2p5_objcon.json: mass and failure - ts_L001_000_linear_h2p5.png: deformed shape coloured by k_xx and σ_xx top, drawn ×11.7 - ts_L001_000_linear_h2p5_hashin.png: Hashin index - The same files for the 5 mm mesh, plus `lam_L001_000.json` and the case files. The `.vtp` files in `results/` open in ParaView.

User · 17:40

Use the shell-visualize tool to plot the z-displacement along the beam

update_plan task list

```
[in_progress] Render uz field with shell-visualize
[pending] Check the image and report
```

```
Plan shown (2 steps).
```

Bash Render z-displacement field, iso and side views

```
PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe; SK=F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills
for v in iso xz; do $PY $SK/shell-visualize/shell_visualize.py results/ts_L001_000_linear_h2p5.npz --field uz --view $v --
png results/ts_L001_000_linear_h2p5_uz_$v.png --no-gui --title "Tape spring L001 [0/0/0], 1 N tip load: u_z" 2>&1 | grep -v
"^Using ASD"; done
```

```
web facets: 0 of 10000 (no web (no catalog boom given))
saved results\ts_L001_000_linear_h2p5.vtp: 10000 facets, 10427 nodes
uz [mm]: min -8.5231, max 1.167e-05
saved results/ts_L001_000_linear_h2p5_uz_iso.png
web facets: 0 of 10000 (no web (no catalog boom given))
saved results\ts_L001_000_linear_h2p5.vtp: 10000 facets, 10427 nodes
uz [mm]: min -8.5231, max 1.167e-05
saved results/ts_L001_000_linear_h2p5_uz_xz.png
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h2p5_uz_iso.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h2p5_uz_xz.png

```
[image]
```

wrap_up wrap-up

```
shell-visualize drew the 2.5 mm mesh coloured by u_z in an iso view and a side (x-z) view. u_z runs from 0 at the clamp to
−8.52 mm at the tip, and the shape is drawn ×11.7.
```

```
Shown in the next-step box. Your closing message is your answer to the user: write it now, in full, as the last thing you
say this turn. The box asks your question, so do not ask it again.
```

Claude · 17:40

I used shell-visualize to draw the boom coloured by z-displacement, from the 2.5 mm mesh results:

- ts_L001_000_linear_h2p5_uz_xz.png: side view (x–z), the clearest view of the deflection along the beam
- ts_L001_000_linear_h2p5_uz_iso.png: iso view, with the undeformed outline in gray

What the plots show: - u_z goes smoothly from 0 at the clamp (black dots) to −8.52 mm at the tip, in the curved shape of a cantilever under a tip load. - The largest positive value is 1.2e-5 mm, which is just round-off at the clamp. - The colour is the same across each cross-section: the section moves down as a whole and does not flatten or twist. This is expected from a linear run.

How to read them: - The shape is drawn 11.7 times larger than the real deflection, so the curvature looks much bigger than it is. The real tip deflection is only 0.85% of the length. - The colour bar is symmetric (±8.52 mm) because the tool always uses one for fields that can have either sign. Every value here is ≤ 0, so only the blue half is used. - In the side view the colour bar covers the tip a little.

shell-visualize only draws the field on the 3D shell; it has no option for a line graph of u_z against x.

User · 17:48

Not now

Claude · 17:48

Okay, no line plot for now. I'm standing by for the next step you want to check.

User · 17:51

Use the opensees-to-abaqus skill to translate the OpenSeesPy script you created above into an Abaqus Python script. Run this in Abaqus. Use the OpenSeesAbaqusResultsComparator agent to compare the Abaqus ODB to the OpenSeesPy results (do not allow the agent to look at the input files to either OpenSeesPy or Abaqus), and report any discrepancies between the two. Only use 16 CPU cores, and do not modify any files do not modify any files on this server outside the F drive / Tara / BenazzoVenkatadri_ThinShellSims folder. Put all code you create during this Abaqus run in Code for Project / 11/11/26 (Tara).

update_plan task list

```
[in_progress] Read opensees-to-abaqus skill and run_abaqus tool
[pending] Translate the OpenSeesPy script to Abaqus
[pending] Run Abaqus (16 CPUs) and extract ODB
[pending] Compare with the comparator agent
[pending] Report discrepancies
```

```
Plan shown (5 steps).
```

Bash Read opensees-to-abaqus skill and run_abaqus tool

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude; cat skills/opensees-to-abaqus/SKILL.md; echo ======; sed -n 1,80p
orchestrator/tools/run_abaqus.py; ls orchestrator/tools
```

```
---
name: opensees-to-abaqus
description: Turns the one standalone OpenSeesPy shell script assembled by the shell-assemble-run skill into an equivalent
Abaqus/CAE Python script (run elsewhere with `abaqus cae noGUI=...`) - geometry, composite or isotropic section, mesh,
supports, rigid ends, loads, linear / Riks / load-control step - that compares its results with the OpenSees results
embedded in it, and checks the Abaqus model against the model the OpenSeesPy script actually builds. Use when the user wants
an Abaqus version of an OpenSeesPy model. Replaces the OpenSeesPyToAbaqusPythonGenerator agent.
---

# opensees-to-abaqus

Script: `opensees_to_abaqus.py` (this folder; uses `../abaqus-export/case_to_abaqus.py`
for the translation and `../regression-tests/abaqus_mock.py` for the check).
Abaqus 2026 is installed on this server: run the generated script ONLY with
[... shortened ...]
```

Bash Read rest of run_abaqus and locate abaqus

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude; sed -n 80,260p orchestrator/tools/run_abaqus.py; which abaqus; where
abaqus 2>/dev/null
```

```
return p.returncode
        except subprocess.TimeoutExpired:
            f.write(f'\nTIMEOUT after {timeout_s} s\n')
            return 'timeout'


def threads_used(msg):
    """'1 HOST x 1 MPI RANK PER HOST x 8 THREADS PER RANK' -> 8 (largest product in the .msg)."""
    best = None
    try:
[... shortened ...]
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 2 of 5 is in progress.
```

Bash Translate the OpenSeesPy script to an Abaqus script

```
cd F:/Tara/BenazzoVenkatadri_ThinShellSims/.claude/skills; PY=/c/Users/tvenkata/miniforge3/envs/shells/python.exe
$PY opensees-to-abaqus/opensees_to_abaqus.py "../../Code for Project/11-11-26
(Tara)/tape_spring_L001_000_tip_load/ts_L001_000_linear_h2p5_opensees.py" 2>&1 | grep -v "^Using ASD"; echo EXIT $?
```

```
OpenSees reference from F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\results\ts_L001_000_linear_h2p5.npz (newer than the script; --rerun to run it again)
wrote F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\abaqus_ts_L001_000_linear_h2p5.py
  1 cross-section piece(s), 400 elements along x, 3 ply/plies ([0/0/0] h = 0.105 mm), 1 support(s), 1 load(s), linear, S4R
  OpenSees reference (ts_L001_000_linear_h2p5.npz): report node u = [0.0, -0.0, -8.523052], max |u_i| = 8.5231 mm, top-ply
stress (s1, s2, t12) = [46.1423, 0.4441, 0.3547] MPa
model check: ts_L001_000_linear_h2p5_opensees.py (recorded OpenSees calls) vs abaqus_ts_L001_000_linear_h2p5.py (stand-in
Abaqus API)
  [PASS] shell nodes = 400 x 25 grid of the case geometry (10426 nodes)  (max distance 0.0e+00 mm)
  [PASS] shell normal of each of the 1 section piece(s) = OpenSees facet normal  (min cos 1.0000)
  [PASS] Abaqus script runs on the stand-in API up to the job
  [PASS] Abaqus seeds: 400 along x, 25 across the section  ([400] along, 25 across)
  [PASS] ply angle of each section the same on every facet (from the
[... shortened ...]
```

Write F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\run_abaqus_in_repo.py

```
"""
Runs .claude/orchestrator/tools/run_abaqus.py unchanged, except that its "only
folder it may write into" (ME295, by default the repo's parent F:\\Tara) is
narrowed to the repo itself, F:\\Tara\\BenazzoVenkatadri_ThinShellSims. The run
folder, Abaqus scratch (TEMP/TMP) and copied results then all stay inside the
repo (11/11/26, Tara: nothing on this server outside that folder may be
modified). The 8-CPU cap of run_abaqus.py is kept.

  python run_abaqus_in_repo.py <dir>/abaqus_<name>.py [--cpus N] [-- <size> <S4R|S4>]
"""
import sys
from pathlib import Path
[... shortened ...]
```

```
File created successfully at: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\run_abaqus_in_repo.py
(file state is current in your context — no need to Read it back)
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 3 of 5 is in progress. App actions available now: commit, create_pr, verify.
```

Bash Run the Abaqus job with 8 CPUs, confined to the repo

```
cd "F:/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)";
/c/Users/tvenkata/miniforge3/envs/shells/python.exe run_abaqus_in_repo.py
tape_spring_L001_000_tip_load/abaqus_ts_L001_000_linear_h2p5.py --cpus 8 2>&1
```

```
Command running in background with ID: bbefsubb6. Output is being written to:
C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\8d2dc832-9cd2-49c7-983d-ffe9975a4847\tasks\bbefsubb6.output.
You will be notified when it completes — do not poll that file or sleep while waiting. Read the file only if the user asks
for progress, or if you need a line it prints before you can go on, such as a server's ready line or port.
Session cwd remains F:\Tara\BenazzoVenkatadri_ThinShellSims\.claude\skills; directory changes made by the backgrounded
command do not apply to subsequent commands.
```

Claude · 17:52

Abaqus is running in the background on 8 CPUs, which is within your 16-core limit. The project tools cap it at 8. I'll compare the results once it finishes.

[17:53] *Background command "Run the Abaqus job with 8 CPUs, confined to the repo" completed (exit code 0)*

Bash Read Abaqus run output and confirm no F:\Tara scratch

```
cat "C:/Users/tvenkata/AppData/Local/Temp/16/claude/67a1dd7b249b42bc/8d2dc832-9cd2-49c7-983d-
ffe9975a4847/tasks/bbefsubb6.output"; ls /f/Tara/
```

```
run_abaqus_in_repo: writes restricted to F:\Tara\BenazzoVenkatadri_ThinShellSims
run folder: F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\abaqus_runs\abaqus_ts_L001_000_linear_h2p5_20261111_175218
Abaqus: C:\SIMULIA\Commands2026\abaqus.BAT
CPUs: 8 (cap 8); scratch: F:\Tara\BenazzoVenkatadri_ThinShellSims\abaqus_scratch\20261111_175218_32904
abaqus cae finished: exit 0, 1.0 min (log F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26
(Tara)\tape_spring_L001_000_tip_load\abaqus_runs\abaqus_ts_L001_000_linear_h2p5_20261111_175218\run_abaqus.log)
job ts_L001_000_linear_h2p5_S4R_2p5mm: COMPLETED; CPUs used 8
extracting ts_L001_000_linear_h2p5_S4R_2p5mm.odb with abaqus python odb_extract.py
  -> written

----- ts_L001_000_linear_h2p5_S4R_2p5mm_results.txt -----
[... shortened ...]
```

update_plan task list

```
{"advance": true}
```

```
Plan updated: step 4 of 5 is in progress.
```

Agent

```
{
 "subagent_type": "OpenSeesAbaqusResultsComparator",
 "description": "Compare OpenSees vs Abaqus results",
 "prompt": "Compare these two result files of the same thin-shell model and report whether they agree, with every
discrepancy quantified:\n\n- OpenSeesPy results: F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26
(Tara)\\tape_spring_L001_000_tip_load\\results\\ts_L001_000_linear_h2p5.npz\n- Abaqus ODB extract:
F:\\Tara\\BenazzoVenkatadri_ThinShellSims\\Code for Project\\11-11-26
(Tara)\\tape_spring_L001_000_tip_load\\ts_L001_000_linear_h2p5_S4R_2p5mm_odb.npz\n\nStrict rules from the user:\n- Use ONLY
these two .npz files as data. Do NOT open, read or grep any model input: no *_opensees.py, abaqus_*.py, *_code/ pieces, case
.json, lam*.json, *_results.txt, *_instructions.txt, abaqus_runs/ contents, or the skills folder. Learn everything you need
from inside the two .npz files.\n- Write every file you create (comparison scripts, plots, report) ONLY into this ne
[... shortened ...]
```

```
Async agent launched successfully. (This tool result is internal metadata — never quote or paste any part of it, including
the agentId below, into a user-facing reply.)
agentId: aabd73b17d1be24ae (internal ID - do not mention to user. Use SendMessage with to: 'aabd73b17d1be24ae', summary:
'<5-10 word recap>' to continue this agent.)
The agent is working in the background. You will be notified automatically when it completes. You know nothing about its
results until that notification arrives — do not report, assume, or predict them; continue other work or respond to the user
in the meantime.
Do not duplicate this agent's work — avoid working with the same files or topics it is using.
output_file: C:\Users\tvenkata\AppData\Local\Temp\16\claude\67a1dd7b249b42bc\8d2dc832-9cd2-49c7-983d-
ffe9975a4847\tasks\aabd73b17d1be24ae.output
Do NOT Read or tail this file via the shell tool — it is the full subagent JSONL transcript and reading it will overflow
your context. If the user asks for progress, say the agent is still running; you'll get a completion notification.
```

Claude · 17:53

The Abaqus job finished in 1 minute on 8 CPUs, and its results file was extracted. The comparator agent is now working from only the two results files (OpenSees and Abaqus), with no access to either model's input files. I'll

report when it finishes.

User · 17:54

As part of your comparison, create plots of z-displacement and stress sigma_xx on the top ply, as you did in OpenSeesPy

ToolSearch


```
{
 "query": "select:SendMessage",
 "max_results": 1
}
```


```
[image]
```

SendMessage


```
{
 "to": "aabd73b17d1be24ae",
 "summary": "Add uz and top-ply sigma_xx 3D plots",
 "message": "Additional plots requested by the user: as part of the comparison, make 3D deformed-shape contour plots (like
PyVista shell plots: deformed shape over the undeformed outline in gray, clamped nodes as black dots, stated magnification,
same magnification and same colour range for both codes) of (1) the z-displacement u_z and (2) the top-surface sigma_xx of
the top ply, each with an OpenSees panel and an Abaqus panel side by side, iso view, symmetric coolwarm colour map with
white = 0. Build them from the two .npz files only (same rules as before: no model input files; write only into the
comparison_opensees_abaqus folder). PyVista is in the same Python env; render offscreen to PNG. Include the PNG paths and
the min/max of each field in your final report.",
 "type": "message",
 "recipient": "aabd73b17d1be24ae",
 "recipient_kind": "agent",
 "content": "Additional plots requested by the user: as p
[... shortened ...]
```



```
{"success":true,"message":"Message queued for delivery to aabd73b17d1be24ae at its next tool round. If it finishes first, it
is resumed with the message and notifies again, so wait for that notification before resending.","pin":
{"id":"aabd73b17d1be24ae","name":"aabd73b17d1be24ae","ref":"60933c"}}
```


Claude · 17:54

I've added the two plots to the comparator's task: z-displacement and top-ply σ_xx, each as an OpenSees panel next to an Abaqus panel. They'll use the same 3D style as the earlier OpenSeesPy plots, with matching magnification and colour ranges. It still works only from the two results files. I'll report when it finishes.

User · 17:58

Status update

Claude · 17:58

The comparator agent is still running and hasn't reported back yet.

Already done: - **Translation:** the OpenSeesPy script is translated to Abaqus, and all 13 model checks pass. - **Abaqus run:** the job (S4R, 2.5 mm mesh, 8 CPUs) completed in about 1 minute. Its results file was extracted to `ts_L001_000_linear_h2p5_S4R_2p5mm_odb.npz`. - **Abaqus's own comparison** (run before the agent):

Quantity	Abaqus	OpenSees	Difference
Tip u_z	−8.583 mm	−8.523 mm	0.70%
Tip rotation	0.012751 rad	0.012662 rad	0.70%
Top-ply σ1	46.33 MPa	46.14 MPa	0.40%
Top-ply σ2	0.464 MPa	0.444 MPa	4.5% (CHECK)
Top-ply τ12	−0.354 MPa	+0.355 MPa	Same size, opposite sign (CHECK)

The two CHECK lines are both small transverse stresses. The τ12 sign flip looks like a shell-normal convention difference: Abaqus printed "shell normal flipped on the face through [0.0, 0.0]". The comparator should confirm whether that's the cause.

The comparator is also making the requested u_z and top-ply σ_xx plots (OpenSees and Abaqus side by side). I'll report the full comparison when it finishes.

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\comparison_opensees_abaqus\pv_uz_deformed.png

```
[image]
```

Read F:\Tara\BenazzoVenkatadri_ThinShellSims\Code for Project\11-11-26 (Tara)\tape_spring_L001_000_tip_load\comparison_opensees_abaqus\pv_sxx_topply_top_deformed.png

```
[image]
```

Bash Check scratch cleanup and changed files

```
ls -la /f/Tara/BenazzoVenkatadri_ThinShellSims/abaqus_scratch 2>&1; cd /f/Tara/BenazzoVenkatadri_ThinShellSims && git status --short | head -20
```

```
total 4
drwxr-xr-x 1 tvenkata 1049089 0 Nov 11 17:53 .
drwxr-xr-x 1 tvenkata 1049089 0 Nov 11 17:52 ..
?? "Code for Project/11-10-26 (Tara)/abaqus_axial_check/r2p5/ctm_axial_r25p62_a90_RIKS.odb"
?? "Code for Project/11-10-26 (Tara)/abaqus_axial_check/r5/ctm_axial_r25p62_a90_RIKS.odb"
?? "Code for Project/11-11-26 (Tara)/run_abaqus_in_repo.py"
?? "Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load/"
?? "Code for Project/11-4-26 (Tara)/system_diagram_verification/"
```

Bash Remove the empty Abaqus scratch folder

```
rmdir /f/Tara/BenazzoVenkatadri_ThinShellSims/abaqus_scratch && echo removed
```

```
removed
```

wrap_up wrap-up

```
Abaqus (S4R, 2.5 mm mesh, 8 CPUs) agrees with OpenSees: displacements, axial strain and fibre stress are within 0.7%. The larger differences sit within about 10 mm of the clamp and tip, and the comparator found no sign or frame problems. The cause of the 0.7% offset is unconfirmed, and nothing is committed.
```

```
Shown in the next-step box. Do not call wrap_up again this turn; the advice that follows is for your next call. recap is two or three sentences (about 300 characters); this one ran 308. Keep it to what is now true and where it stands. Your closing message is your answer to the user: write it now, in full, as the last thing you say this turn. The box asks your question, so do not ask it again.
```

Claude · 18:00

The Abaqus and OpenSeesPy models agree. Displacements, rotations, axial strains and fibre-direction ply stress are within 0.7% everywhere. The comparator found no sign, axis or shell-normal problems; it worked only from the two results files and never opened either model's input files.

What was run - **Translation:** the opensees-to-abaqus skill turned the 2.5 mm OpenSeesPy script into an Abaqus script, and all 13 model checks pass. - **Abaqus run:** S4R elements on the same 2.5 mm mesh, 8 CPUs, finished in 1 minute. - You allowed up to 16 cores, but the project's run tool caps Abaqus at 8. - Abaqus scratch files would normally go to `F:\Tara\abaqus_scratch`, outside the folder you allowed. I ran the tool through a small wrapper, `Code for Project/11-11-26 (Tara)/run_abaqus_in_repo.py`, which keeps everything inside `BenazzoVenkatadri_ThinShellSims`. - The scratch folder was emptied after the run and I removed it. Abaqus's own license client may still keep logs under the user profile; I can't control that.

Key values

Quantity	OpenSees	Abaqus	Difference
Tip u_z	−8.523 mm	−8.583 mm	0.70%
Tip rotation	0.012662 rad	0.012751 rad	0.70%
σ1 (fibre direction), top ply, top surface	46.14 MPa	46.33 MPa	0.4%

Quantity	OpenSees	Abaqus	Difference
σ_xx range, top ply	−29.6 to 46.1 MPa	−29.8 to 46.3 MPa	under 1%
τ12, top ply	0.355 MPa	0.354 MPa	0.1%

The opposite τ12 sign in the Abaqus run log was a labelling problem in that quick check. In the full comparison both normals point the same way and the signs agree.

Differences found 1. **Abaqus is 0.70% more flexible everywhere, by the same factor along the whole beam.** - The comparator suggests the OpenSees in-plane stiffness A11 is 0.7% too high. I doubt that: the earlier laminate-properties check measured A directly from an OpenSees element and found it matches laminate theory to about 1e-17. - My guess, not tested: the strains saved by OpenSees may leave out the element's internal enhanced strains (ASDShellQ4 keeps them for a single-angle laminate). It may also be an element difference, since the OpenSees mesh isn't fully converged (8.596 mm at 5 mm, 8.523 mm at 2.5 mm). - A 1.25 mm run of both codes would settle which. 2. **Larger differences within about 10 mm of the clamp and the tip.** - These are in the transverse and twisting curvatures, their moments, and the in-plane rotation: up to +32% in κ_yy, −67% in κ_xy and M12, +40% in M22. - Away from the ends these fields agree within about 1–2%. - These quantities are about 1% of the main values, and the difference is typical of the two element types at a clamp corner. 3. **The small cross-section stresses σ2 and N22 differ by 5–7%.** These fields are close to zero, and Abaqus S4R shows element-to-element noise near the root. This is noise, not a model difference.

Not compared: reactions and transverse shear forces are not in either results file. The comparator checked equilibrium cross-section by cross-section instead: Abaqus matches the applied moment exactly, OpenSees is 0.7% low, which is the same 0.70% offset.

The plots you asked for are in `Code for Project/11-11-26 (Tara)/tape_spring_L001_000_tip_load/comparison_opensees_abaqus/`, each with OpenSees and Abaqus side by side at the same magnification (×20) and colour range. In both, the shapes and colour patterns look the same. - pv_uz_deformed.png: z-displacement - pv_sxx_topply_top_deformed.png: σ_xx on the top surface of the top ply

The folder also has 10 more comparison plots, the comparator's script and log, and a comparison report.

User · 18:39

Create a PDF in the Claude Conversations folder with the contents of this conversation

update_plan task list

```
[in_progress] Find the Claude Conversations folder
[pending] Write the conversation content
[pending] Build and check the PDF
```

```
Plan shown (3 steps).
```

Bash Find conversations folder and check PDF libraries

```
cd /f/Tara/BenazzoVenkatadri_ThinShellSims; find . -maxdepth 3 -iname "*claude conv*" -not -path "./.git/*"; ls "Claude
Conversations" 2>/dev/null | head; /c/Users/tvenkata/miniforge3/envs/shells/python.exe -c "import reportlab;
print('reportlab', reportlab.Version)" 2>&1; python -c "import reportlab; print('sys reportlab')" 2>&1 | tail -1
```

```
./Claude Conversations
Federico 11-5-26 Conversation 1 (Orchestrator v3.0 - protocol, design screen, study manager, hook).pdf
Tara 10-23-26 Conversation 1 (Python setup and Federico's 10-22 test cases).pdf
Tara 10-23-26 Conversation 2 (Building material property agents).pdf
Tara 10-23-26 Conversation 3 (Testing material property agents).pdf
Tara 10-26-26 Conversation 1 (CFRP laminate test cases, result viewer and Abaqus agents).pdf
Tara 11-10-26 Conversation 1 (Attempt at baseline with Palmeri and Laurenzi results).pdf
Tara 11-11-26 Conversation 1 (Baseline with tools only, no orchestrator).pdf
Tara 11-11-26 Conversation 2 (Optimization to replicate baseline).pdf
Tara 11-4-26 Conversation 1 (Skill to join laminate, mesh and BC-load OpenSeesPy code into one script, GitHub sync).pdf
[... shortened ...]
```

Bash Look for existing PDF generator and PDF libraries

```
cd /f/Tara/BenazzoVenkatadri_ThinShellSims; grep -rl "Claude Conversations" --include=*.py . 2>/dev/null | head; for p in
python /c/Users/tvenkata/miniforge3/python.exe /c/Users/tvenkata/miniforge3/envs/shells/python.exe; do echo "== $p"; $p -c "
import importlib
for m in ['reportlab','fpdf','pypdf','markdown','weasyprint','fitz','docx2pdf']:
    try: importlib.import_module(m); print('have',m)
    except Exception: pass" 2>&1; done; which pandoc wkhtmltopdf msedge 2>&1 | grep -v "no "; ls "/c/Program Files
(x86)/Microsoft/Edge/Application/msedge.exe" 2>&1
```

```
./Code for Project/11-11-26 (Tara)/conversation_to_pdf.py
== python
== /c/Users/tvenkata/miniforge3/python.exe
== /c/Users/tvenkata/miniforge3/envs/shells/python.exe
/c/Program Files (x86)/Microsoft/Edge/Application/msedge.exe
```

Bash Read the existing conversation-to-PDF script

```
cd "/f/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; ls; sed -n 1,200p conversation_to_pdf.py
```

```
(.*?) L001_000.json
L001_909090.json
SUMMARY.md
TapeSpring_L001_000_report.pdf
_pylibs
ab
abaqus_scratch
build_report.py
coil_hub_check.py
conversation_to_pdf.py
[... shortened ...]
```

Bash Locate this session's transcript file

```
ls -la /c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/8d2dc832-9cd2-49c7-983d-ffe9975a4847.jsonl
```

```
-rw-r--r-- 1 tvenkata 1049089 4623924 Nov 11 18:40 /c/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-
ThinShellSims/8d2dc832-9cd2-49c7-983d-ffe9975a4847.jsonl
```

update_plan task list

```
{"done": [1, 2], "start": 3}
```

```
Plan updated: step 3 of 3 is in progress.
```

Bash Convert this session's transcript to PDF

```
cd "/f/Tara/BenazzoVenkatadri_ThinShellSims/Code for Project/11-11-26 (Tara)"; python conversation_to_pdf.py
"C:/Users/tvenkata/.claude/projects/F--Tara-BenazzoVenkatadri-ThinShellSims/8d2dc832-9cd2-49c7-983d-ffe9975a4847.jsonl"
"F:/Tara/BenazzoVenkatadri_ThinShellSims/Claude Conversations/Tara 11-11-26 Conversation 3 (System diagram verification -
laminate code, tape spring model and Abaqus comparison).pdf" "Tara 11-11-26 Conversation 3: System diagram verification -
laminate code, tape spring model and Abaqus comparison" 2>&1 | tail -3
```